

Name: MODEL SOLUTIONS.

Student Number:

Please answer ALL questions.

This test is worth 25% of your overall class grade.
Each problem statement has the associated mark value stated.

Do not turn over your cover sheet to start until you are told to begin. Extra blank pages are given for each problem with what should be enough room to answer each question, especially if you use both sides. However, if you need extra paper, it will be provided.

If you separate the pages, ensure that they are stapled together again when handed in.

There is a total of 40 marks (19 marks for Question 1 and 21 marks for Question 2)

90 minutes total time allowed (plus 5-10 minutes to upload your code).

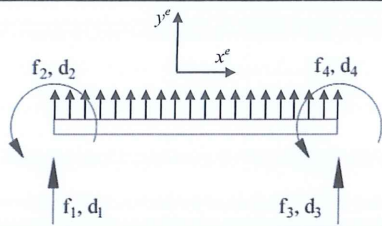
Circle or box all final answers! You may use pencil but make sure to highlight and/or ink final answers.

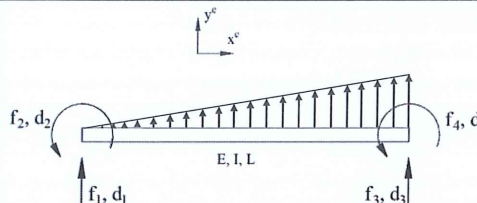
Label each part of each question, include appropriate units, and an appropriate number of significant figures, for all answers.

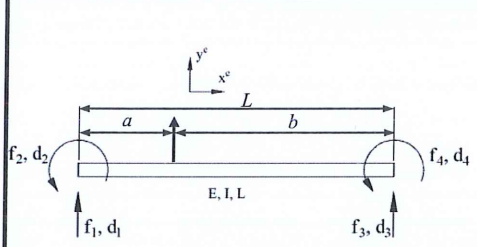
Some equations for equivalent nodal loading are presented on the next page for your reference.

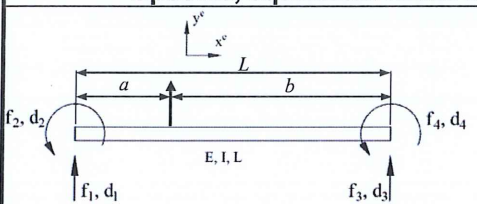
At the completion of the test, you are asked to upload the code that you used to complete the test via the link on the course Ako | Learn page. While there are no marks allocated for your code, having access to your code helps with the awarding of partial credit when final answers are incorrect.

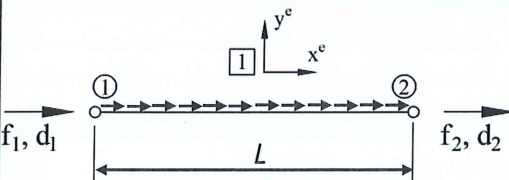
The equivalent nodal loading definitions (in element co-ordinates) are provided for reference.

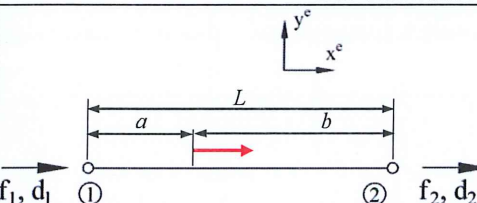
Equivalent nodal loading definition for a Uniformly Distributed Load (UDL)		
Beam Element Formulation		Frame Element Formulation
	$f_{eq,UDL}^e = \begin{Bmatrix} \bar{w}L/2 \\ \bar{w}L^2/12 \\ \bar{w}L/2 \\ -\bar{w}L^2/12 \end{Bmatrix}$	$f_{eq,UDL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ \bar{w}L/2 \\ \bar{w}L^2/12 \\ 0 \\ \bar{w}L/2 \\ -\bar{w}L^2/12 \end{Bmatrix}$

Equivalent nodal loading definition for a Linearly Varying Load (LVL)		
Beam Element Formulation		Frame Element Formulation
	$f_{eq,LVL}^e = \begin{Bmatrix} 3\bar{w}L/20 \\ \bar{w}L^2/30 \\ 7\bar{w}L/20 \\ -\bar{w}L^2/20 \end{Bmatrix}$	$f_{eq,LVL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ 3\bar{w}L/20 \\ \bar{w}L^2/30 \\ 0 \\ 7\bar{w}L/20 \\ -\bar{w}L^2/20 \end{Bmatrix}$

Equivalent nodal loading definition for a transverse point load		
Beam Element Formulation		Frame Element Formulation
	$f_{eq,PL}^e = \bar{w} \begin{Bmatrix} 1 - \frac{3a^2}{L^2} + \frac{2a^3}{L^3} \\ \frac{a^3}{L^2} - \frac{2a^2}{L} + a \\ \frac{3a^2}{L^2} - \frac{2a^3}{L^3} \\ \frac{a^3}{L^2} - \frac{a^2}{L} \end{Bmatrix}$	$f_{eq,PL}^e = \bar{w} \begin{Bmatrix} 0 \\ 1 - 3(a^2/L^2) + 2(a^3/L^3) \\ (a^3/L^2) - 2(a^2/L) + a \\ 0 \\ 3(a^2/L^2) - 2(a^3/L^3) \\ (a^3/L^2) - (a^2/L) \end{Bmatrix}$

Specific, equivalent nodal loading definition for a mid-span point load, where $a = L/2$		
Beam Element Formulation		Frame Element Formulation
	$f_{eq,PL}^e = \begin{Bmatrix} \bar{w}/2 \\ \bar{w}L/8 \\ \bar{w}/2 \\ -\bar{w}L/8 \end{Bmatrix}$	$f_{eq,PL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ \bar{w}/2 \\ \bar{w}L/8 \\ 0 \\ \bar{w}/2 \\ -\bar{w}L/8 \end{Bmatrix}$

Equivalent nodal loading definition for a Distributed Axial Load		
Bar Element Formulation		Frame Element Formulation
	$f_{eq,axial}^e = \bar{p} \begin{Bmatrix} L/2 \\ L/2 \end{Bmatrix}$	$f_{eq,axial}^e = \bar{p} \begin{Bmatrix} L/2 \\ 0 \\ 0 \\ L/2 \\ 0 \\ 0 \end{Bmatrix}$

Equivalent nodal loading definition for a Concentrated Axial Load		
Bar Element Formulation		Frame Element Formulation
	$f_{eq,axial\ PL}^e = \bar{p} \begin{Bmatrix} 1 - a/L \\ a/L \end{Bmatrix}$	$f_{eq,axial\ PL}^e = \bar{p} \begin{Bmatrix} 1 - a/L \\ 0 \\ 0 \\ a/L \\ 0 \\ 0 \end{Bmatrix}$

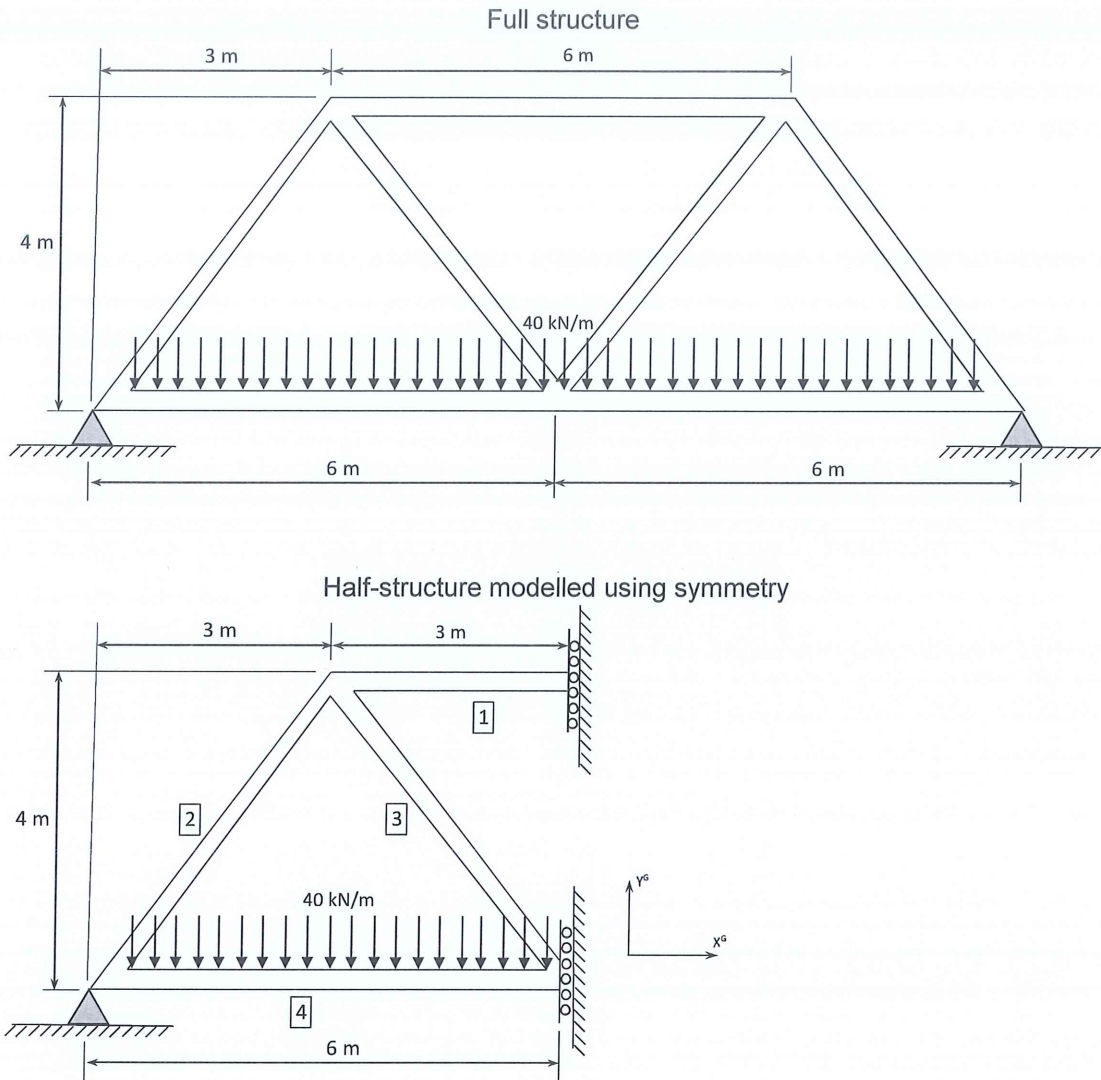
Problem 1: A bridge with simple supports (pins) is modelled carrying a Uniformly Distributed Load (UDL) that represents the bridge deck weight is shown in the diagram on the next page. The magnitude of the Uniformly Distributed Load intensity is 40kN/m.

We will simplify the problem by using symmetry and only model the left half of the bridge. The boundary conditions that we apply at the middle of the structure (at the plane of symmetry) are rollers which prevent rotation and horizontal translation, but do not provide any vertical reaction force. The half-structure that we will model is made up of four frame elements and has a simple (pin) support at the bottom left corner.

Assume that all elements are frame elements, made of steel, and that each element has the same cross-sectional area and second moment of area, with values given on the next page. The length of the elements differ (as given on the next page), and the structure itself is considered mass-less relative to the loads it carries.

[19 Total Marks]

- a. Draw a sketch of the structure showing the global degrees of freedom $\mathbf{q}$ for the model. **[2 Marks]**
- b. Draw a free-body diagram for each element, showing the element degrees of freedom in global co-ordinates as well as the orientation of your element co-ordinate system (including stating the transformation angle, α , for each element). **[2 Marks]**
- c. Write out the assembly matrices $\mathbf{A}^e$ for each element **[2 Marks]**
- d. Write out the element equivalent forcing terms for element 4 ($\mathbf{f}_{eq}^4$) to represent the Uniformly Distributed Load. Present the overall global forcing vector, $\mathbf{Q}$. **[2 Marks]**
- e. Solve for the overall deflections of the system, $\mathbf{q}$, and present your result. **[1 Mark]**
- f. Sketch (by hand) the original and deflected shape of the bridge structure, indicating the direction and magnitude of all deflection components (displacements and rotations). The sketch does not need to be to scale, but each component of deflection must be labelled. **[3 Marks]**
- g. Provide the reaction components (forces and moments) at all supports, parallel and perpendicular to the supports. Provide a sketch of these reaction loads, showing the magnitude and direction of all reaction components. **[3 Marks]**
- h. Determine the maximum bending moment magnitude (positive or negative) induced within element 2 (M_{max}^2) and state where it occurs within the element. **[2 Marks]**
- i. If the bridge deck is not allowed to deflect more than 70mm at the centre of the full structure, what is the maximum intensity of UDL that can be applied to the bridge deck to ensure that displacement does not exceed this limit? **[2 Marks]**



All elements are FRAME elements

Elastic Modulus: $E^1 = E^2 = E^3 = E^4 = 200 \text{ GPa}$

Second Moment of Area: $I^1 = I^2 = I^3 = I^4 = 6 \times 10^{-6} \text{ m}^4$

Cross-sectional Area: $A^1 = A^2 = A^3 = A^4 = 2 \times 10^{-4} \text{ m}^2$

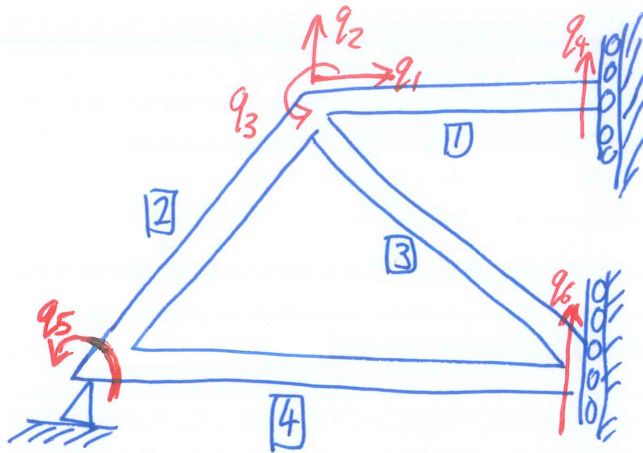
Lengths: $L^1 = 3 \text{ m}$

$L^2 = 5 \text{ m}$

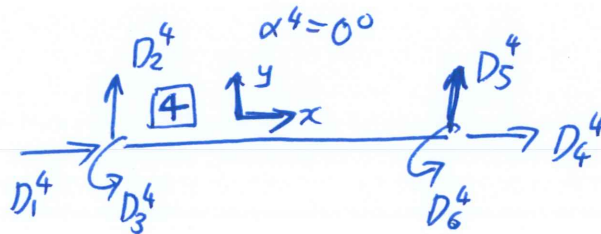
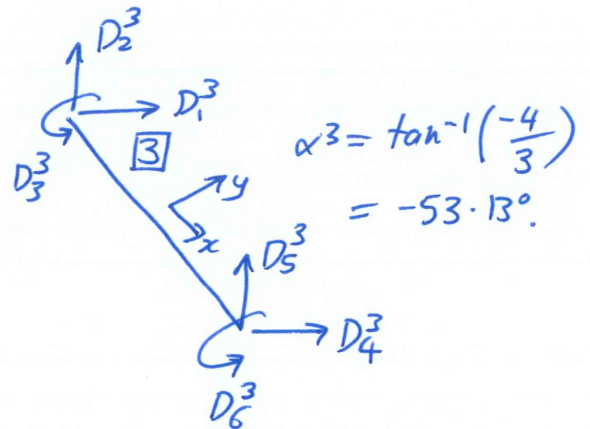
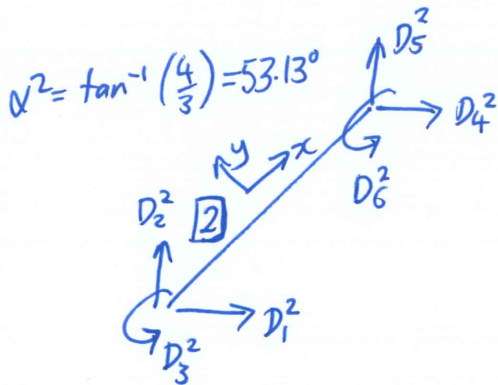
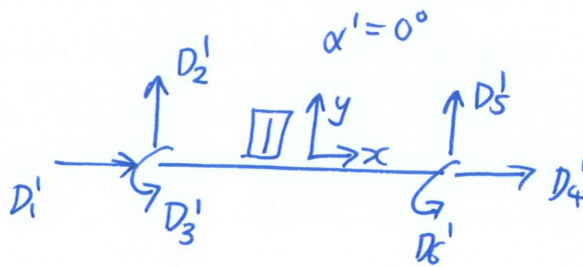
$L^3 = 5 \text{ m}$

$L^4 = 6 \text{ m}$

a)



b)



$$c) \quad A^1 = \begin{matrix} & D_1^1 & D_2^1 & D_3^1 & D_4^1 & D_5^1 & D_6^1 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$A^2 = \begin{matrix} & D_1^2 & D_2^2 & D_3^2 & D_4^2 & D_5^2 & D_6^2 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$A^3 = \begin{matrix} & D_1^3 & D_2^3 & D_3^3 & D_4^3 & D_5^3 & D_6^3 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$A^4 = \begin{matrix} & D_1^4 & D_2^4 & D_3^4 & D_4^4 & D_5^4 & D_6^4 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

d) $f_{eq, udc}^4 = \begin{Bmatrix} 0 \\ \bar{w}L/2 \\ \bar{w}L^2/12 \\ 0 \\ \bar{w}L/2 \\ -\bar{w}L^2/12 \end{Bmatrix} = \begin{Bmatrix} 0 \\ (-40,000)(6)/2 \\ (-40,000)(6^2)/12 \\ 0 \\ (-40,000)(6)/2 \\ -(-40,000)(6^2)/12 \end{Bmatrix} = \begin{Bmatrix} 0 \text{ kN} \\ -120 \text{ kN} \\ -120 \text{ kNm} \\ 0 \text{ kN} \\ -120 \text{ kN} \\ +120 \text{ kNm} \end{Bmatrix}$

$\bar{w} = -40 \text{ kN/m}$

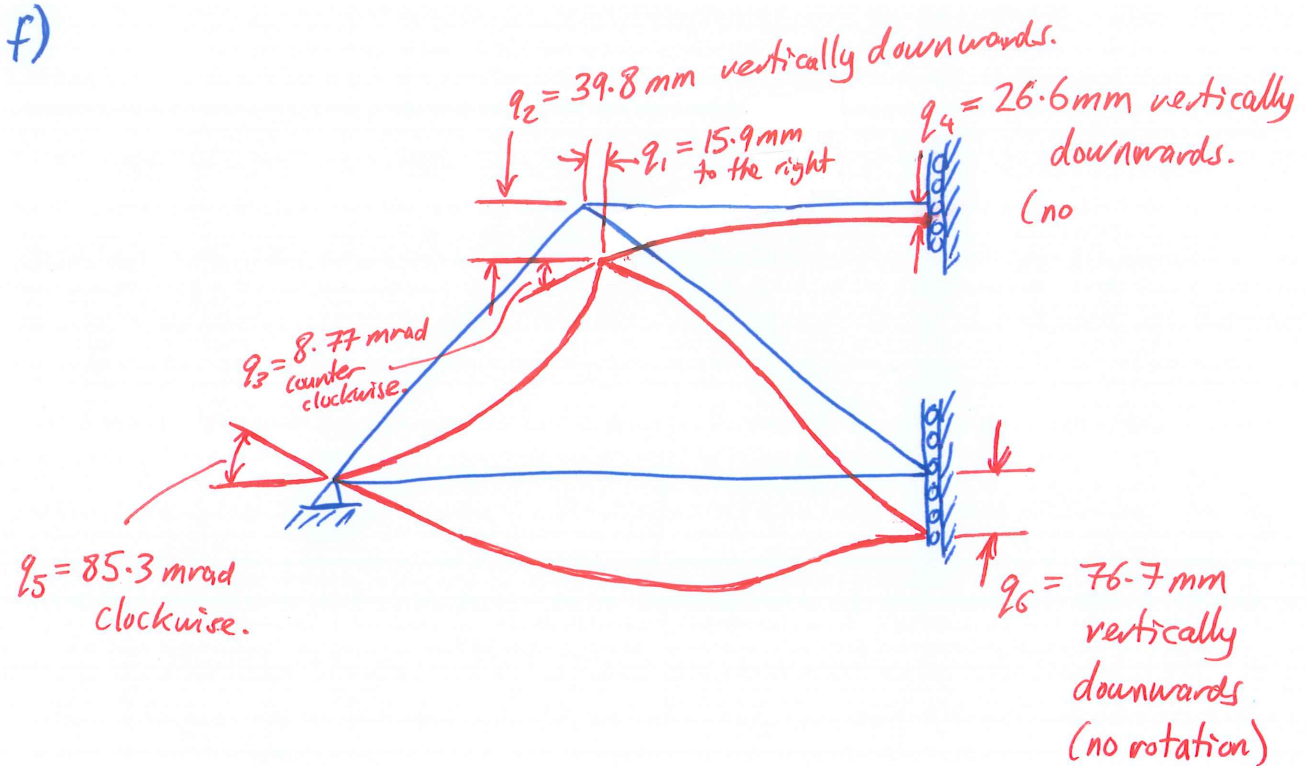
$$F_{eq, udc}^4 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}^T f_{eq, udc}^4 = \begin{Bmatrix} 0 \\ -120,000 \\ -120,000 \\ 0 \\ -120,000 \\ +120,000 \end{Bmatrix} = \begin{Bmatrix} 0 \text{ kN} \\ -120 \text{ kN} \\ -120 \text{ kNm} \\ 0 \text{ kN} \\ -120 \text{ kN} \\ +120 \text{ kNm} \end{Bmatrix}$$

$\mathbf{A}^4 = \mathbf{I}_{6 \times 6}$ as $\alpha^4 = 0^\circ$.

$$Q = Q_{eq, udc}^4 = \begin{Bmatrix} 0 \text{ kN} \\ 0 \text{ kN} \\ 0 \text{ kNm} \\ 0 \text{ kN} \\ -120 \text{ kNm} \\ -120 \text{ kN} \end{Bmatrix}$$

e)

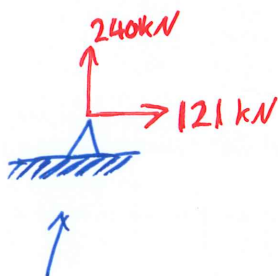
$$q = \begin{Bmatrix} 0.0159 \text{ m} \\ -0.0398 \text{ m} \\ 0.0088 \text{ rad} \\ -0.0266 \text{ m} \\ -0.0853 \text{ rad} \\ -0.0767 \text{ m} \end{Bmatrix} = \begin{Bmatrix} 15.9004 \text{ mm} \\ -39.7833 \text{ mm} \\ 8.7721 \text{ mrad} \\ -26.6252 \text{ mm} \\ -85.2789 \text{ mrad} \\ -76.7116 \text{ mm} \end{Bmatrix}$$



g) Support Reactions:

3.51 kNm
 212 kN

$$F'_{4-6} = \begin{Bmatrix} -212 \text{ kN} \\ 0 \text{ kN} \\ -3.5088 \text{ kNm} \end{Bmatrix}$$



$$F_{1-3}^2 + F_{1-3}^4 + (F_{eq, u04, 1-3}^4)$$

$$= \begin{Bmatrix} 121.2297 \text{ kN} \\ 131.9417 \text{ kN} \\ -67.1192 \text{ kNm} \end{Bmatrix} + \begin{Bmatrix} 0 \text{ kN} \\ -11.9417 \text{ kN} \\ -52.8808 \text{ kNm} \end{Bmatrix} - \begin{Bmatrix} 0 \\ -120 \text{ kN} \\ -120 \text{ kNm} \end{Bmatrix} = \begin{Bmatrix} 121.2297 \text{ kN} \\ 240 \text{ kN} \\ 0 \text{ kNm} \end{Bmatrix}$$

$$F_{4-6}^3 + F_{4-6}^4 + (-F_{eq, u04, 4-6}^4)$$

124 kNm
 90.8 kN

$$= \begin{Bmatrix} 90.7760 \text{ kN} \\ -131.9417 \text{ kN} \\ 14.2553 \text{ kNm} \end{Bmatrix} + \begin{Bmatrix} 0 \text{ kN} \\ 11.9417 \text{ kN} \\ -18.7693 \text{ kNm} \end{Bmatrix} - \begin{Bmatrix} 0 \\ -120 \text{ kN} \\ +120 \text{ kNm} \end{Bmatrix} = \begin{Bmatrix} 90.7760 \text{ kN} \\ 0 \text{ kN} \\ -124.514 \text{ kNm} \end{Bmatrix}$$

h)

$$f^2 = \begin{Bmatrix} 178.2912 \text{ kN} \\ -17.8188 \text{ kN} \\ -67.1192 \text{ kNm} \\ -178.2912 \text{ kN} \\ 17.8188 \text{ kN} \\ -21.9747 \text{ kNm} \end{Bmatrix}$$

The larger magnitude of the two moment terms is 67.2 kNm and it occurs at the lower end of the element, at the pin support.

- NOTE: - This element carries a constant shear force.
- Shear force is equal to the rate of change of bending moment.
 - Therefore, the bending moment within the element will vary linearly from the value at node ① to the value at node ②.
 - The maximum bending moment anywhere within the element will not exceed the values at each end.

i) This is a linear elastic analysis and the UDL is the only load acting on the structure.

Therefore, vertical deflection will scale linearly with UDL intensity.

When a UDL of intensity 40 kN/m was applied, the vertical deflection in the centre was 76.7 mm

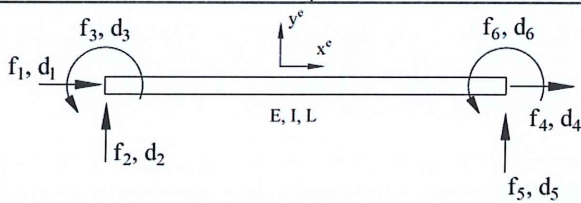
$$\text{Allowable UDL} = \frac{70}{76.7} \times 40 = 36.5 \text{ kN/m}$$

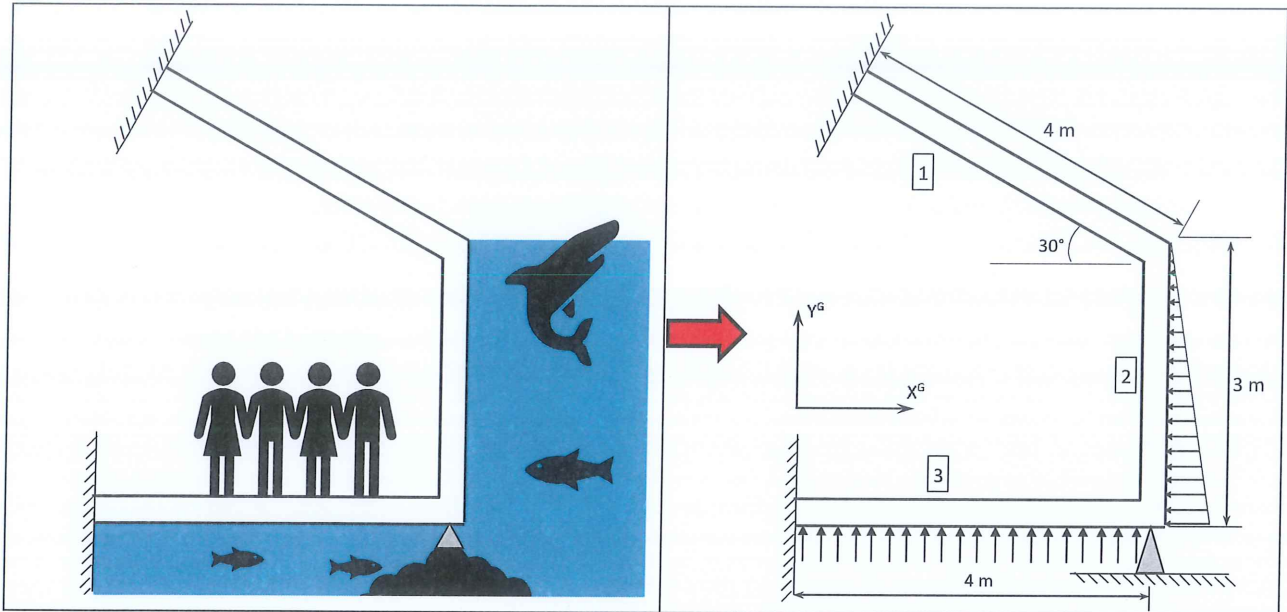
Problem 2: An aquarium provides an underwater viewing platform for people to see aquatic life. The structure is made up of three frame elements, has fixed supports at the top left and bottom left corner, and is anchored to a rocky outcrop on the sea floor via a simple support (pin) in the bottom right corner.

Assume that all three elements are frame elements, made of steel, with the same cross-sectional area and second moment of area ($E = 200 \text{ GPa}$, $I = 1 \times 10^{-4} \text{ m}^4$, and $A = 2 \times 10^{-4} \text{ m}^2$). The length of each element differs as shown on the diagram, and the structure itself is considered mass-less relative to the loads it carries.

[21 Total Marks]

- Draw a sketch of the structure showing the global degrees of freedom $\mathbf{q}$. **[2 Marks]**
- Draw the free body diagram of each element, showing the element degree of freedom in global coordinates and the orientation of your element co-ordinate system (including stating the transformation angle, α , for each element). **[2 Marks]**
- Write out the assembly matrices $\mathbf{A}^e$ for each element **[2 Marks]**
- Assume that the water has a density of $1,000 \text{ kg/m}^3$, the acceleration due to gravity is 9.81 N/kg , and that the frame being modelled supports a section of wall (the distance into the page) of 4m .
Write out the element equivalent forcing terms for both elements 2 and 3 ($\mathbf{f}_{eq}^2$ and $\mathbf{f}_{eq}^3$) to represent the water pressure. Present the overall global forcing vector, $\mathbf{Q}$. If you cannot calculate the peak intensity of the linearly varying load (which is also the intensity of the uniformly distributed load), use a magnitude of $\bar{w}$ of 100 kN/m .
Note: The formulae on the second page of this test may be a useful reference. **[4 Marks]**
- Solve for the overall deflections of the system, $\mathbf{q}$, and present your result. **[1 Mark]**
- Clearly sketch the overall deflected shape of the system.
→ In your sketch, show all the deflections and rotations that were calculated and label the magnitudes. The sketch does not need to be to scale, but each component must be labelled. **[3 Marks]**
- Provide the support reaction forces and moments at all three support points, parallel and perpendicular to the supports. Provide a sketch for each case, showing the magnitude and direction of all reaction components. **[3 Marks]**
- Determine the axial force, axial stress, and axial normal strain in element 1 (from axial loads only, ignoring any bending effects), and indicate whether they are tensile or compressive. **[2 Marks]**
- Present the element displacement vector (in element co-ordinates) for element 1 and determine the total deflection (in the X^G and Y^G directions) of a point located at a distance of 3m (along the element) from the top of element 1. The shape functions (applied to the frame element numbering system) are given below. Include your working. **[2 Marks]**

List of the shape functions for a frame element.			
$\psi_1(x) = \left(1 - \frac{x}{L}\right); \psi_2(x) = \frac{x}{L}$			
$N_1(x) = 1 - \frac{3x^2}{L^2} + \frac{2x^3}{L^3}$	$N_2(x) = \frac{x^3}{L^2} - \frac{2x^2}{L} + x$	$N_3(x) = \frac{3x^2}{L^2} - \frac{2x^3}{L^3}$	$N_4(x) = \frac{x^3}{L^2} - \frac{x^2}{L}$
		$u(x) = \psi_1(x)d_1 + \psi_2(x)d_4$ $v(x) = N d^e = \begin{pmatrix} N_1(x) & N_2(x) & N_3(x) & N_4(x) \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix}$	



All elements are FRAME elements

Elastic Modulus: $E^1 = E^2 = E^3 = 200 \text{ GPa}$

Second Moment of Area: $I^1 = I^2 = I^3 = 1 \times 10^{-4} \text{ m}^4$

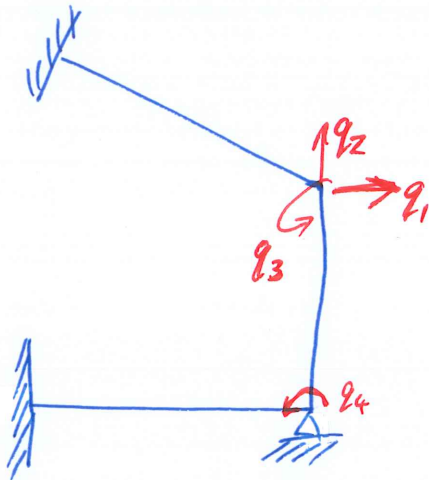
Cross-sectional Area: $A^1 = A^2 = A^3 = 2 \times 10^{-4} \text{ m}^2$

Lengths: $L^1 = 4 \text{ m}$

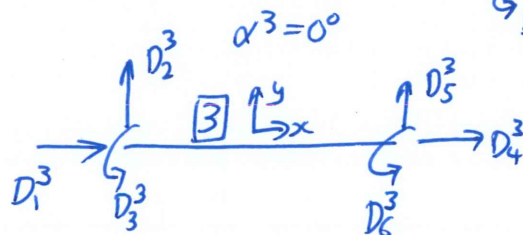
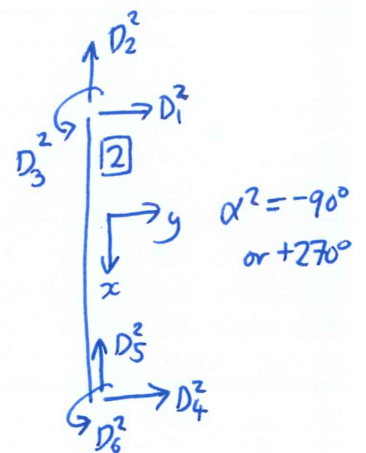
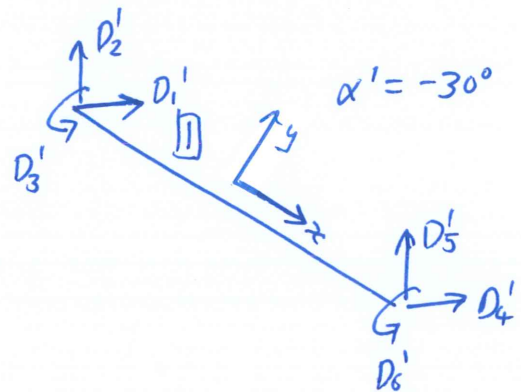
$L^2 = 3 \text{ m}$

$L^3 = 4 \text{ m}$

a)



b)



c)

$$A^1 = \begin{matrix} & D_1^1 & D_2^1 & D_3^1 & D_4^1 & D_5^1 & D_6^1 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$A^2 = \begin{matrix} & D_1^2 & D_2^2 & D_3^2 & D_4^2 & D_5^2 & D_6^2 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$A^3 = \begin{matrix} & D_1^3 & D_2^3 & D_3^3 & D_4^3 & D_5^3 & D_6^3 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

d) $|\bar{w}| = \rho g h d = (1,000)(9.81)(3)(4) = 117,720 \text{ N/m}.$

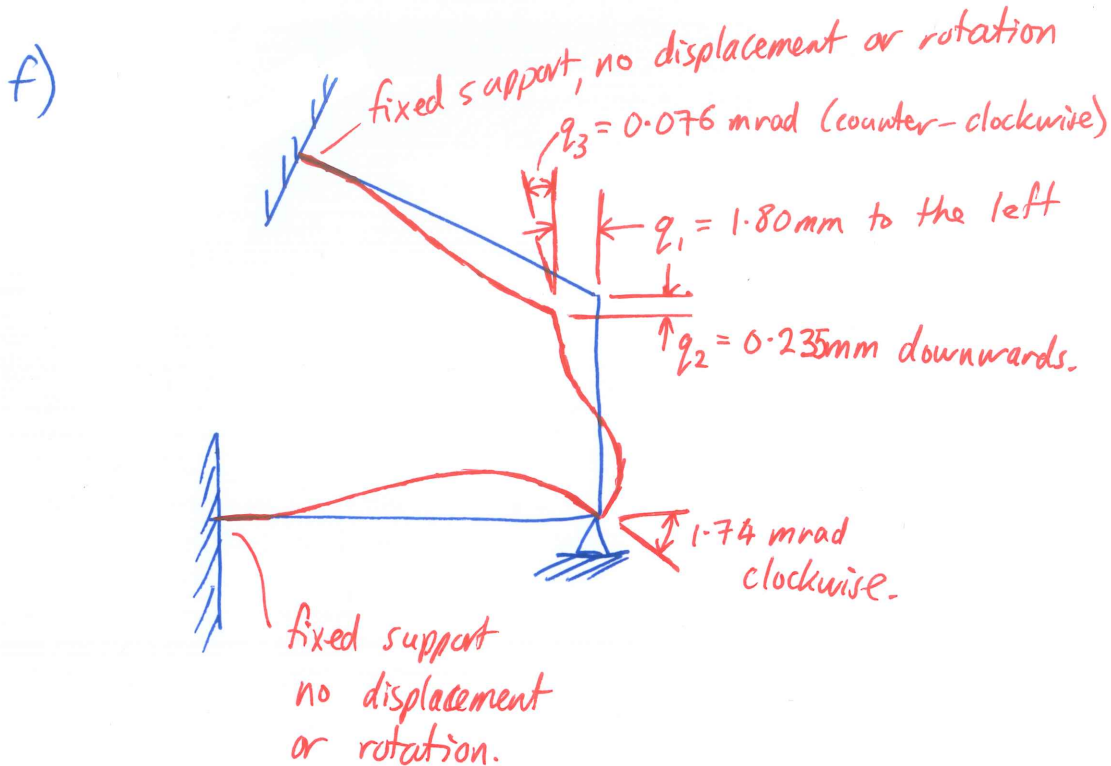
$$f_{eq, LVL}^2 = \begin{Bmatrix} 0 \\ 3\bar{w}L/20 \\ \bar{w}L^2/30 \\ 0 \\ 7\bar{w}L/20 \\ -\bar{w}L^2/20 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 3(-117,720)(3)/20 \\ (-117,720)(3^2)/30 \\ 0 \\ 7(-117,720)(3)/20 \\ -(-117,720)(3^2)/20 \end{Bmatrix} = \begin{Bmatrix} 0 \text{ N} \\ -52,974 \text{ N} \\ -35,316 \text{ Nm} \\ 0 \text{ N} \\ -123,606 \text{ N} \\ 52,974 \text{ Nm} \end{Bmatrix}$$

$$f_{eq, UDL}^3 = \begin{Bmatrix} 0 \\ \bar{w}L/2 \\ \bar{w}L^2/12 \\ 0 \\ \bar{w}L/2 \\ -\bar{w}L^2/12 \end{Bmatrix} = \begin{Bmatrix} 0 \\ (117,720)(4)/2 \\ (117,720)(4^2)/12 \\ 0 \\ (117,720)(4)/2 \\ -(117,720)(4^2)/12 \end{Bmatrix} = \begin{Bmatrix} 0 \text{ N} \\ 235,440 \text{ N} \\ 156,960 \text{ Nm} \\ 0 \text{ N} \\ 235,440 \text{ N} \\ -156,960 \text{ Nm} \end{Bmatrix}$$

$$Q = Q_{eq, LVL}^2 + Q_{eq, UDL}^2 = \begin{Bmatrix} -52,974 \text{ N} \\ 0 \text{ N} \\ -35,316 \text{ Nm} \\ 52,974 \text{ Nm} \end{Bmatrix} + \begin{Bmatrix} 0 \text{ N} \\ 0 \text{ N} \\ 0 \text{ Nm} \\ -156,960 \text{ Nm} \end{Bmatrix} = \begin{Bmatrix} -52,974 \text{ N} \\ 0 \text{ N} \\ -35,316 \text{ Nm} \\ -103,986 \text{ Nm} \end{Bmatrix}$$

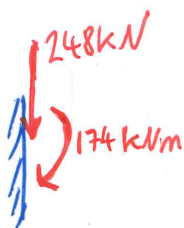
e).

$$q = \begin{Bmatrix} -0.0018 \text{ m} \\ -0.0002 \text{ m} \\ 0.0001 \text{ rad} \\ -0.0017 \text{ rad} \end{Bmatrix} = \begin{Bmatrix} -1.8007 \text{ mm} \\ -0.2348 \text{ mm} \\ 0.0762 \text{ mrad} \\ -1.7356 \text{ mrad} \end{Bmatrix}$$



g)

$$f'_{1-3} = \begin{Bmatrix} 14.42 \text{ kN} \\ 4.7103 \text{ kN} \\ 9.0396 \text{ kNm} \end{Bmatrix}$$



$$219 \text{ kN} \quad F_{46}^2 + F_{46}^3 + (-F_{q_1, \text{LVK}}^2) + (-F_{q_2, \text{LVK}}^3)$$

$$= \begin{Bmatrix} 38.1307 \\ 3.1308 \\ -69.2746 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 13.0168 \\ -34.7114 \end{Bmatrix} - \begin{Bmatrix} -123.6060 \\ 0 \\ 52.9740 \end{Bmatrix} - \begin{Bmatrix} 0 \\ 235.440 \\ -156.966 \end{Bmatrix}$$

$$= \begin{Bmatrix} 161.7367 \text{ kN} \\ -219.2924 \text{ kN} \\ 0 \text{ kNm} \end{Bmatrix}$$

$$F_{1-3}^3 + (-F_{q_1, \text{LVK}}^3) = \begin{Bmatrix} 0 \\ -13.0168 \\ -17.3597 \end{Bmatrix} - \begin{Bmatrix} 0 \\ 235.440 \\ +156.966 \end{Bmatrix} = \begin{Bmatrix} 0 \text{ kN} \\ -248.4568 \text{ kN} \\ -174.3157 \text{ kNm} \end{Bmatrix}$$

h)

$$f' = \begin{Bmatrix} 14.42 \text{ kN} \\ 4.71 \text{ kN} \\ 9.04 \text{ kNm} \\ -14.42 \text{ kN} \\ -4.71 \text{ kN} \\ 9.80 \text{ kNm} \end{Bmatrix}$$

$$d' = \begin{Bmatrix} 0 \text{ mm} \\ 0 \text{ mm} \\ 0 \text{ mm} \\ -1.4420 \text{ mm} \\ -1.1037 \text{ mm} \\ 0.0762 \text{ mm} \end{Bmatrix}$$

axial force = 14.4 kN compression.

$$\begin{aligned} \text{axial normal strain} &= \frac{\Delta L}{L} = \frac{(-1.4420 \times 10^{-3}) - (0)}{4} = -3.605 \times 10^{-4} \epsilon \\ &= -360.5 \mu\epsilon \end{aligned}$$

axial strain has a magnitude of 361 $\mu\epsilon$ and is compressive.

axial stress:

$$\sigma = \frac{F}{A} = \frac{14,420}{2 \times 10^{-4}} = 72.1 \text{ MPa in compression.}$$

or:

$$\sigma = E\epsilon = (200 \times 10^9)(-360.5 \times 10^{-6}) = -72.1 \text{ MPa.}$$

axial normal stress from axial loading only is 72.1 MPa.
in compression.

$$i) \quad d' = \begin{Bmatrix} 0 \text{ mm} \\ 0 \text{ mm} \\ 0 \text{ mrad} \\ -1.4420 \text{ mm} \\ -1.1037 \text{ mm} \\ 0.0762 \text{ mrad} \end{Bmatrix}$$

$$u(x) = \psi_1(x) \overset{0}{d_1} + \psi_2(x) d_4$$

$\psi_1(x)$ not needed
as it is multiplied
by zero anyway.

$$\psi_2(x) = \frac{x}{L} = \frac{3}{4} = 0.75.$$

$$u(x) = \psi_2(x) d_4 = (0.75)(-1.4420) \\ = -1.0815 \text{ mm}.$$

$$v(x) = \overset{0}{N_1(x)} d_2 + \overset{0}{N_2(x)} d_3 + N_3(x) d_5 + N_4(x) d_6$$

$N_1(x)$ and $N_2(x)$ not needed
as they will be multiplied
by zero anyway.

$$N_3(x) = \frac{3x^2}{L^2} - \frac{2x^3}{L^3}$$

$$N_3(3) = \frac{3(3)^2}{4^2} - \frac{2(3)^3}{4^3} = 0.84375.$$

$$N_4(x) = \frac{x^3}{L^2} - \frac{x^2}{L}$$

$$N_4(3) = \frac{3^3}{4^2} - \frac{3^2}{4} = -0.5625.$$

$$v(x) = N_3(x) d_5 + N_4(x) d_6 \\ = (0.84375)(-1.1037) + (-0.5625)(0.0762) \\ = -0.9741 \text{ mm}.$$

$$\delta X^G = u(x) \cos(\alpha') - v(x) \sin(\alpha') \\ = (-1.0815) \cos(-30^\circ) - (-0.9741) \sin(-30^\circ) \\ = -1.424 \text{ mm}$$

$$\delta Y^G = u(x) \sin(\alpha') + v(x) \cos(\alpha') \\ = (-1.0815) \sin(-30^\circ) + (-0.9741) \cos(-30^\circ) \\ = -0.3028 \text{ mm}$$

The point 3m along element [1] (from the top support)
deflects -1.42mm in the X^G direction
and -0.303mm in the Y^G direction.